

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	Valid method e.g. $B = \mu_0 n I$ to calculate B or I or n or μ_0 (1) Gives $B = 0.0606$ [T] or $I = 7.13$ [A] or $n = 26\,972$ or $\mu_0 = 2.51 \times 10^{-6}$ (1) Correct conclusion based on calculation (with ecf) e.g. Lindsay wrong (if B or I calculated), Lindsay wrong because μ_0 too large or real value is 1.26×10^{-6} or Lindsay wrong because n value too large or real value is 13 500 (allow ecf on $n = 27\,000$ here) (1) i.e. Calculating $B = 0.121$ [T] followed by Lindsay correct is 2 marks (1 st and 3 rd marks)			3	3	1	
		(ii)	Iron / ferromagnetic / steel core Accept increase permeability Don't accept increase permeability of free space	1			1		
	(b)	(i)	Use of $BAN \cos \theta$ (1) Correct answer = 0.046 Wb or Wb turn or T m ² **unit mark** (1)	1	1		2	1	
		(ii)	Max when $\theta = 90$ because flux changing / cutting at max rate (accept cutting most lines or similar for max rate) (1) When $\theta = 0$ no emf because flux max or no cutting or rate of change is zero (1) Alternative for 2nd mark Min when $\theta = -90, 270$ because flux changing / cutting in opposite direction (1) Allow 1 mark if both angles are correct with no explanation Allow 1 mark if angles are wrong but both explanations are correct (must refer to flux changing or cutting but no need to insist on "rate")		2		2		
			Question 1 total	2	3	3	8	2	0